Limitations and claim boundaries I

The boundaries here define the contribution. The central goal is to show — concretely and testably — that the categorical generalized-variational construction inspired by FedGVI (Mildner et al. 2025) has standard-Bayes client limits and a project-local log-linear-pool server identity. Under the explicit bridge in sec. 5.8, the latter specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol; behavior away from those limits is evaluated only under declared simulation conditions. The exact identities are formal; performance and deployment claims remain conditional. Items beyond that boundary are named as future work.

Three robustness axes: theorem, heuristic, and objective I

Robustness enters in three places with unequal standing — a distinction surfaced by an adversarial review of this work and preserved deliberately throughout (sec. 3.3).

Three robustness axes: theorem, heuristic, and objective I

1. **Client-side, source-theorem-backed.** A robust per-agent generalized-Bayes update with a bounded loss — β-loss (Basu et al. 1998) or generalized cross-entropy (Zhang and Sabuncu 2018) — inside the generalized posterior. Density-power and gamma-divergence Bayes make clear that this is a loss/divergence-specific robustness property, not an automatic property of every generalized posterior (Fujisawa and Eguchi 2008; Ghosh and Basu 2015). It is derived from the generalized-Bayes objective (Bissiri et al. 2016; Jewson et al. 2018; Knoblauch et al. 2022), provably limits to NLL/Bayes (hence to the standard pool) as the loss parameter goes to zero (eq. 4, eq. 5), and is the theorem-bearing axis under the matching loss, divergence, and regularity assumptions of FedGVI. The manuscript does not re-prove that theorem for every possible categorical data-generating process. **This is the client-side mechanism implemented and evaluated here.**

Three robustness axes: theorem, heuristic, and objective II

2. **Server-side, heuristic.** The divergence-reweighting aggregator `robust_aggregate`. Its positive formal property is the recovery limit — robustness 0 equals the standard log-linear pool (eq. 7). A scoped no-go rejects one declared separable objective class; it is *not* a closed-form minimizer of a FedGVI objective, and we never claim it inherits the bounded-influence bound. An empirical characterization (sec. 14.2) makes this boundary concrete: the heuristic has a *finite* measured breakdown point (a colluding majority captures it), a witness against unconditional server resistance rather than a transferred client theorem. The robustness sweep reports it as a complementary heuristic, and fig. 13 is labeled accordingly. Robust federated optimizers such as Krum and gamma-mean aggregation motivate the adversarial client problem (Blanchard et al. 2017; Li et al. 2022), but they do not certify this belief-pooling heuristic. It has BH-rejected positive contrasts in the configured accuracy verdict in sec. 7.5.1, while the declared mechanism gallery and conditional-world study retain reversals that prohibit a universal accuracy claim.

Three robustness axes: theorem, heuristic, and objective III

3. **Server-side, objective-backed (conservative).** The variational aggregator `variational_aggregate` of sec. 5.8.2, derived in sec. 11. It runs exact closed-form block updates that descend the stated free energy eq. 8 monotonically on each block, shares the recovery corner (eq. 7) with axis 2, and — unlike axis 2 — carries a proven raw effective-weight bound with empirical redescending behavior (fig. 15). Its cost in the declared comparison is conservatism: the entropy regularization yields a more diffuse consensus, and the method does *not* win that configured peak-accuracy verdict. It closes the "is there *any* objective-backed server rule with raw-weight control" gap; it does not retroactively endow the sharp axis-2 heuristic with an objective.

The honest state is therefore a triangle, not a binary: axis 1 is source-theorem-backed under matching assumptions, axis 2 has conditional wins and reversals *without* a server-side objective, and axis 3 is objective-backed *but* conservative. No experiment attributes axis 2's accuracy win to a theoretical guarantee, and none attributes axis 3's weight bound to a peak-accuracy claim. The remaining open problem — an objective whose minimizer is the *sharp* reweighting itself, which would combine accuracy and guarantee in one server rule — is named future work in sec. 8.17; the extended methods that broaden the toolkit without touching these claims are cataloged in sec. 12.

Scope boundaries that the evidence does not cross I

Scope boundaries that the evidence does not cross I

- **The bridge is a recasting, not an upstream claim.** Friston et al. (Friston et al. 2024) supply a belief-sharing operator for agents with a shared world model; Mildner et al. (Mildner et al. 2025) supply robust federated generalized variational inference. Neither paper claims the other. The contribution here is to recast the former as the tested KL/NLL zero-robustness recovery corner of the latter-inspired construction, then measure what changes when robust losses and server reweighting are introduced.
- **Historical sources are conceptual, not formal support.** The pre-modern sources added in sec. 8.9 support a genealogy of expectation, inverse inference, utility, and collective judgment (Huygens 1657; Bayes 1763; Laplace 1774; Bernoulli 1738; Condorcet 1785). They do not support any claim about KL, NLL, product-of-experts training, FedGVI, or `robust_aggregate`. Those claims rest only on the modern cited formalism and the tests reported in sec. 7.1.

Scope boundaries that the evidence does not cross II

- **Single-machine federation.** Federation is validated with queue transport, a single-machine OS-process helper, and loopback TCP: agents serialize beliefs, the server aggregates, and consensus is broadcast back without changing the mathematics. The socket path now exercises frame integrity and file-backed digest-verified replay validation; an optional SQLite round-ID guard also survives local process restarts. These controls are still not a substitute for cross-host deployment, identity-bound mTLS, a shared multi-host replay domain, discovery, long-running worker orchestration, or fault tolerance; those steps are scoped as future work (sec. 8.17).

Scope boundaries that the evidence does not cross III

- **GPU / PyTorch Bayesian-neural-network experiments at the original FedGVI scale.** The bounded-influence result is anchored here by a small NumPy logistic-regression baseline (fig. 16), not the RTX-class runs of the source paper (Mildner et al. 2025). Even on that anchor the robust client's margin over the standard client is non-monotone rather than uniform: it opens in the moderate-to-high contamination range and then vanishes at the most extreme swept level, where both configurations collapse together with no reliable ordering (sec. 7.6). That terminal convergence is reported, not trimmed; the axis's rigorous standing rests on the recovery identities and the FedGVI theorem under its matching assumptions, not on the size or monotonicity of the empirical gap in this figure.

Scope boundaries that the evidence does not cross IV

- **Classification baselines are point estimates, not full posteriors.** The NumPy logistic-regression baseline and the PyTorch MLP complement both use point-estimate weights (no posterior covariance is computed). A genuine mean-field variational family over the weights — diagonal-Gaussian $q(w)$ with a closed-form KL and a Monte-Carlo ELBO — is implemented as a tested primitive (`bnn_variational_torch.VariationalMLP`, recovering the point-estimate net exactly as its posterior variance vanishes), but the full paper-faithful FedGVI classification lane (that variational family trained under the contamination sweep, at GPU scale) is not: MCMC, non-diagonal structure, and stochastic-weight averaging remain unimplemented, and the full-posterior regime remains unverified.
- **Hierarchical depth does not by itself improve the base task.** The two- and three-level stacks (sec. 13.9, sec. 13.11) run alternating L1/L2(/L3) inference end-to-end and resolve their added context latents above chance, but on the shared location task they do not beat the flat baseline — the paired location-accuracy gap is a small, statistically reliable negative gap at the reported seed count (fig. 25). Depth is therefore validated as executable and consensus-preserving, not as an accuracy improvement; whether a richer policy-and-horizon task family rewards hierarchy is future work (sec. 8.20).

Scope boundaries that the evidence does not cross V

- **Discrete POMDP only — continuous or hybrid state spaces are not addressed.** This work is discrete-categorical only, matching the community's worked POMDP example (Da Costa et al. 2020; Friston et al. 2024). The limit-as-proof contract is validated off the categorical case only for a one-dimensional Gaussian-mean conjugate slice; continuous-state active inference and Gaussian belief-sharing colonies remain untested.
- **Temperature and divergence calibration are fixed, not learned.** Coarsened posterior and safe-Bayes theory make clear that the learning-rate/temperature is part of the inference rule under misspecification (Miller and Dunson 2018; Grünwald 2012; Kleijn and Vaart 2012). This manuscript validates the configured losses and divergences, but it does not learn an optimal coarsening radius, temperature, or divergence schedule across agents.
- **Real multi-machine federation, networking, and privacy cryptography.** Federation here is *mathematical* (factor aggregation), not infrastructural — unlike communication-efficient or Byzantine-robust federated learning (McMahan et al. 2017; Blanchard et al. 2017).
- **New linguistic theory.** The language-acquisition study (sec. 7.3) reproduces the Dirichlet count mechanism (eq. 14) mechanically; it does not extend the linguistics.

What the statistics can and cannot claim I

The verdict is a paired comparison at a single high contamination rate (sec. 7.5.1, tbl. 5). This concentrates power where the effect is largest, which is honest about *where* robustness pays off but does not characterize the full contamination curve as a continuous function; the per-rate table (tbl. 4) reports the rest of the sweep without claiming family-wide significance beyond what BH-FDR (Benjamini and Hochberg 1995) certifies. BH-FDR controls expected false discovery proportion within the declared family, not the chance of any false positive. The matched-pairs Wilcoxon tests are rank tests under paired-difference assumptions (Wilcoxon 1945; Fay and Proschan 2010), not assumption-free proofs about raw means. Confidence intervals are percentile bootstrap (Efron and Tibshirani 1993), not analytic, and inherit that method's small-sample caveats at the lowest trial counts.

What the statistics can and cannot claim I

The power values are observed-effect design approximations useful for confirmatory sample-size planning; they are not independent evidence for the verdict and do not cover model-specification uncertainty outside the seeded simulation harness (Wasserstein and Lazar 2016).